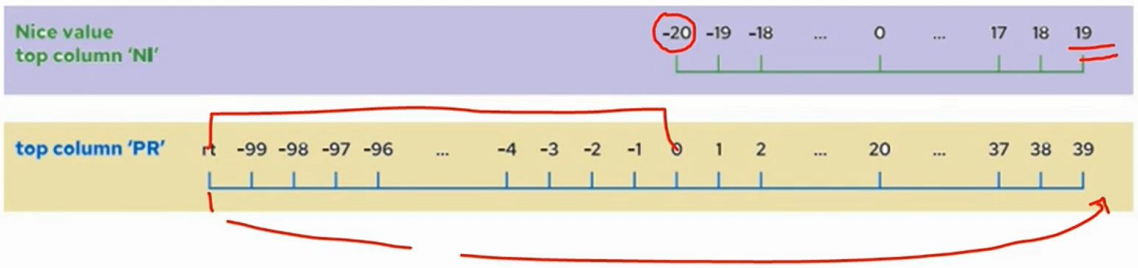
ABDUL JANEEB K

INFULENCE PROCESS SCHEDULING

- Suppose we created a process
- We creating a process(job) sleep 500
- When we entering top we can get the process running below and see also there priority
- We can influence a process's CPU scheduling priority using the nice and renice commands
- Instead of we can use nicevalue
- Total priority range -100 to 39
- -100 to 0 is reserved for system process.
- Nice value range -20 to 19
- 0 is highest priority and 39 is lowest priority
- -20 is the highest cpu scheduling priority and 19 Is the lowest cpu scheduling priority

Process scheduling in Unix-like operating systems involves determining the order and priority in which processes are executed by the CPU. The nice and renice commands are tools that influence this process scheduling by allowing users to adjust the priority of their processes.



Total Priority Range (PR): The total priority range in the top command spans from -100 to 39.

Reserved Range: The range from -100 to 0 is reserved for system processes. These processes usually have higher priority than user processes.

User Process Range: The range from 0 to 39 is typically for user processes.

Priority Values: The priority value is what determines the actual priority of a process. It ranges from 0 (highest priority) to 39 (lowest priority).

Nice Value: The nice value is used to adjust the priority of a process. It ranges from -20 to 19. A lower nice value corresponds to a higher priority.

root@Astra:~

```
[root@Astra ~]# nice -n -20 sleep 500 &
[2] 7415
[1]   Done                    nice -n 10 sleep 500
[root@Astra ~]# ps
    PID TTY          TIME CMD
   6960 pts/2    00:00:00 bash
   7415 pts/2    00:00:00 sleep
   7416 pts/2    00:00:00 ps
[root@Astra ~]# ps -o pid,ni,pr,comm -p 7415
error: unknown user-defined format specifier "pr"

Usage:
 ps [options]

 Try 'ps --help <simple|list|output|threads|misc|all>'
  or 'ps --help <s|l|o|t|m|a>'
 for additional help text.

For more details see ps(1).
[root@Astra ~]# ps -o pid,ni,pri,comm -p 7415
    PID  NI PRI COMMAND
   7415 -20  39 sleep
[root@Astra ~]# ^C
[root@Astra ~]# renice -n 10 -p 7415
7415 (process ID) old priority -20, new priority 10
[root@Astra ~]# ps -o pid,ni,pri,comm -p 7415
    PID  NI PRI COMMAND
   7415  10   9 sleep
[root@Astra ~]#
```